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Inventor(s)

Jiang; Zelian

FOLDING CART

Abstract

A folding cart comprises a front frame, a rear frame, a side frame arranged between the front frame and the rear frame and used for expanding and folding, and a release mechanism used for restricting or releasing restriction on the folding of the side frame. A force direction applied to trigger the release mechanism is the same as that applied to fold the side frame, the side frame is driven to fold after the release mechanism releases the restriction on the folding of the side frame. In the folding cart, after an external force is applied to trigger the release mechanism to release the folding restriction, the external force in the same direction can be continuously applied to directly drive the side frame to fold. A user only needs to apply force in a single direction to complete dual tasks of unlocking and folding one after another.

Inventors: Jiang; Zelian (Zhongshan, CN)

Applicant: Zhongshan Kanglaiya Trade Co., Ltd. (Zhongshan, CN)

Family ID: 1000008629713

Assignee: Zhongshan Kanglaiya Trade Co., Ltd. (Zhongshan, CN)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The application claims priority to Chinese patent application No. 202520516166.2, filed on Mar. 24, 2025, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present invention belongs to the technical field of carts, and particularly relates to a folding cart.

BACKGROUND

[0003] In daily life, as common transportation tools, carts are widely used in various scenarios, such as shopping, transportation of goods and infant outings. However, folding mechanisms of traditional carts typically use independent buttons to release the folding restrictions. When folding the cart, a user needs to find and press the unlock button first, which is typically positioned in specific positions of the cart, such as a side of a frame or near a handle. After the unlock button is pressed, the folding restriction of the cart is released, the user needs to apply another force in another direction to fold the cart together simultaneously, but in this way, the user needs to apply two forces in different directions simultaneously: one is to press the unlock button and another is to fold the cart. The different directions of these forces increase the operational complexity, which makes the operational process more difficult and inconvenient for user. Additionally, since the two directional forces must be applied simultaneously, the user often needs both hands when folding the cart, making one-handed operation impossible, which significantly limits the usage convenience of the cart.

SUMMARY

[0004] To solve the problem that the cart in the existing technology needs to be applied two forces in different directions simultaneously in the folding operation, resulting in difficulty and inconvenience, the present invention provides a folding cart.

[0005] The present invention is realized through the following technical solutions.

[0006] A folding cart comprises a front frame, a rear frame, a side frame for expanding and folding arranged between the front frame and the rear frame, and a release mechanism used for restricting or releasing restriction on the folding of the side frame. A force direction applied to trigger the release mechanism to release restriction is the same as that applied to fold the side frame, after the release mechanism releases restriction on the folding of the side frame, the side frame is driven to fold.

[0007] The folding cart as described above, the side frame includes a front top rod and a front bottom rod whose front ends are pivotally connected to an upper side and a lower side of the front frame respectively, a rear top rod and a rear bottom rod whose rear ends are pivotally connected to an upper side and a lower side of the rear frame respectively, a front inclined rod whose front end is pivotally connected to the lower side of the front frame, a rear inclined rod whose rear end is pivotally connected to the lower side of the rear frame, and a pivoting mechanism which is respectively connected with a rear end of the front top rod, a front end of the rear top rod, a rear end of the front inclined rod and a front end of the rear inclined rod, and enables the four ends to be pivotally connected to each other. A rear end of the front bottom rod is pivotally connected to a front end of the rear bottom rod.

[0008] The folding cart as described above, the release mechanism includes a front bottom pivot member provided at the rear end of the front bottom rod, the front end of the rear bottom rod is

provided with a rear bottom pivot seat pivotally connected to a middle portion of the front bottom pivot member, and a connection pivot pipe arranged between the two adjacent front bottom pivot members. The connection pivot pipe is internally provided with lock pins respectively corresponding to the connection pivot pipes on both sides, and an outer side of the connection pivot pipe is sleeved with sliding sleeves connected to the lock pins. The rear bottom rod is provided with a lock hole for an insertion of the lock pins to restrict the folding of the side frame, and an elastic member is provided between the two lock pins. When the folding cart is in the expanded state, the front bottom pivot member abuts against the rear bottom rod, and the elastic member drives the lock pin to insert into the lock hole.

[0009] The folding cart as described above, the release mechanism further includes driving parts connected to the lock pins through a pulling wire, the connection pivot pipe is provided with sliding grooves for the driving parts to pass through and connect to the sliding sleeves, and a pulling member connecting the two sliding sleeves.

[0010] The folding cart as described above, when the pulling member is pulled upward, both of the sliding sleeves approach each other under the action of force, driving the two driving parts inside the connection pivot pipe to pull out the corresponding lock pins on both sides from the lock holes, thus to release the restriction on the folding of the side frame. Simultaneously, the side frame folds and drives the front frame and the rear frame to approach each other, thereby completing the folding of the folding cart.

[0011] The folding cart as described above, the pivoting mechanism is pivotally connected to the front top rod through a front top pivot seat, a front side of the front top pivot seat is pivotally connected to the rear end of the front top rod and a rear side of the front top pivot seat is pivotally connected to the pivoting mechanism. The pivoting mechanism is pivotally connected to the rear top rod through a rear top pivot seat, a rear side of the rear top pivot seat is pivotally connected to the front end of the rear top rod and a front side of the rear top pivot seat is pivotally connected to the pivoting mechanism.

[0012] The folding cart as described above, the side frame further includes a locking mechanism for keeping the folding cart in a folded state when the folding cart is in the folded state, the locking mechanism includes a folding buckle and a folding hook fastened on the folding buckle. The folding buckle is provided on the front inclined rod and the folding hook is provided on the rear inclined rod; or the folding buckle is provided on the rear inclined rod and the folding hook is provided on the front inclined rod.

[0013] The folding cart as described above, a slope face is arranged on an end of the lock pin close to the lock hole.

[0014] The folding cart as described above, the connection pivot pipe has a prismatic shape, and the driving part and the lock pin are correspondingly configured as a prismatic shape.

[0015] The folding cart as described above, the front top pivot seat is provided with a front support rod extending downward to be pivotally connected to the front bottom rod, and the rear top pivot seat is provided with a rear support rod extending downward to be pivotally connected to the rear bottom rod.

[0016] Compared with the prior art, the present invention has the following advantages.

[0017] In the folding cart according to the present invention, after an external force is applied to trigger the release mechanism to release restriction to release the folding restriction, the external force in the same direction can be continuously applied to directly drive the side frame to fold. The user only needs to apply force in a single direction to complete the dual tasks of unlocking and folding one after another, without additionally applying forces in other directions to drive the cart to fold.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0018] In order to more clearly illustrate the technical solutions in the embodiments of the present invention, the drawings needed in the descriptions of the embodiments will be briefly introduced hereafter. Apparently, the following described drawings are merely a part of the embodiments of the present invention, other drawings can be obtained by those of ordinary skill in the art according to these drawings without any creative works.

[0019] FIG. 1 is a front view of a folding cart in an expanded state according to an embodiment of the present invention.

[0020] FIG. 2 is a three-dimensional view of FIG. 1.

[0021] FIG. 3 is a partially exploded view of FIG. 2.

[0022] FIG. 4 is an enlarged view of Part A in FIG. 3.

[0023] FIG. 5 is a front view of the folding cart in a folded state according to the embodiment of the present invention.

[0024] FIG. 6 is a top view of FIG. 1.

[0025] FIG. 7 is a schematic cross-sectional view of line A-A in FIG. 6 when it's restricted.

[0026] FIG. 8 is a schematic cross-sectional view of line A-A in FIG. 6 when restriction is released.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0027] In order to make the technical problems, technical solutions and beneficial effects solved by the present invention more clearly, the present invention is further illustrated hereafter in combination with the drawings and embodiments. It should be understood that the embodiments described here are only used to explain the present invention but not to limit the present invention.

[0028] Please refer to FIGS. 1-8, a folding cart comprises a front frame 1, a rear frame 2, a side frame 3 arranged between the front frame 1 and the rear frame 2 and used for expanding and folding, and a release mechanism 4 used for restricting or releasing restriction on the folding of the side frame 3. A force direction applied to trigger the release mechanism 4 to release restriction is the same as that applied to fold the side frame 3, after the release mechanism 4 releases the restriction on the side frame, the side frame 3 is driven to fold.

[0029] According to the folding cart of the present invention, after an external force is applied to trigger the release mechanism to release the folding restriction, the external force in the same direction can be continuously applied to directly drive the side frame to fold. A user only needs to apply force in a single direction to complete the dual tasks of unlocking and folding one after another, without additionally applying forces in other directions to drive the cart to fold.

[0030] Further, as the preferred implementation method of this scheme rather than limited, the side frame 3 includes a front top rod 31 and a front bottom rod 32 whose front ends are pivotally connected to upper and lower sides of the front frame 1, respectively, a rear top rod 33 and a rear bottom rod 34 whose rear ends are pivotally connected to upper and lower sides of the rear frame 2, respectively, a front inclined rod 35 whose front end is pivotally connected to the lower side of the front frame 1, a rear inclined rod 36 whose rear end is pivotally connected to the lower side of the rear frame 2, and a pivoting mechanism 37 which is respectively connected with a rear end of the front top rod 31, a front end of the rear top rod 33, a rear end of the front inclined rod 35 and a front end of the rear inclined rod 36, and enables the four ends to be pivotally connected to each other. A rear end of the front bottom rod 32 is pivotally connected to a front end of the rear bottom rod 34.

[0031] In this embodiment, the design of the side frame 3 forms a flexible and stable structure by skillfully connecting and pivoting the front top rod 31, the front bottom rod 32, the rear top rod 33, the rear bottom rod 34, the front diagonal rod 35, and the rear diagonal rod 36, so that the folding cart can keep steady and smooth in the process of expanding and folding, ensuring that the side frame 3 can stably support the whole cart in the expanding state. The configuration of the front diagonal rod 35 and the rear diagonal rod 36 not only enhances the stability of the cart, but also provides the essential guidance and support during the folding process. As a critical connection

point, the pivoting mechanism **37** pivotally interconnects the front top rod **31**, the rear top rod **33**, the front diagonal rod **35** and the rear diagonal rod **36**, so that these rods can move together to realize the expanding and folding of the cart. Additionally, the rear end of the front bottom rod **32** is pivotally connected to the front end of the rear bottom rod **34** to further enhance the stability and flexibility of the side frame **3**. When the cart needs to be folded, the user triggers the release mechanism **4** to release the folding restriction on the side frame **3**, and then applies a force in one direction, and the force is transmitted through the pivotal relationship of these rods, and thus guiding the side frame **3** to fold in an orderly manner. Since all the rods are pivotally connected, they can be folded together in a coordinated manner without jamming or interference.

[0032] Further, as the preferred implementation method of this scheme rather than limited, the release mechanism **4** includes a front bottom pivot member **41** provided at the rear end of the front bottom rod **32**. The front end of the rear bottom rod **34** is provided with a rear bottom pivot seat **42** pivotally connected to a middle portion of the front bottom pivot member **41**, and a connection pivot pipe **43** arranged between the two adjacent front bottom pivot members **41**. The connection pivot pipe **43** is internally provided with lock pins **44** respectively corresponding to the connection pivot pipes **43** on both sides, and an outer side of the connection pivot pipe **43** is sleeved with sliding sleeves **45** connected to the lock pins **44**; the rear bottom rod **34** is provided with lock holes **46** for the insertion of the lock pins **44** to restrict the folding of the side frame **3**, and an elastic member **47** is provided between the two lock pins **44**. When the folding cart is in the expanded state, the front bottom pivot member abuts against the rear bottom rod, and the elastic member drives the lock pins to insert into the lock hole.

[0033] In this embodiment, precise control of the folding state of the side frame **3** is achieved, which provides users with a convenient and reliable operation experience. The front bottom pivot member **41** is provided on the rear end of the front bottom rod **32** and is pivotally connected to the rear bottom pivot seat **42** at the front end of the rear bottom rod **34**, and this connection manner ensures the stability of the side frame **3** in the unfolded state. The connection pivot pipe **43** is provided between the two adjacent front bottom pivot members **41**, and the lock pins **44** are provided inside the connection pivot pipe **43** on both sides. These lock pins **44** can automatically insert into the lock holes **46** on the rear bottom rod **34** under the action of elastic member **47** (such as spring, etc.), so as to lock the side frame **3** and prevent it from accidentally folding in the unfolded state. The sliding sleeves **45** are sleeved on outer sides the connection pivot pipes **43** and connected with the locking pins **44**. When the user needs to fold the cart, by operating the sliding sleeves **45**, the elastic force of the elastic member **47** can be overcome, so that the elastic member is compressed, and the lock pin **44** is pulled out from the lock hole **46** to release the lock of the side frame **3**. Since the sliding sleeve **45** is directly connected to the lock pin **44**, the user only needs to make a simple sliding motion to complete the unlocking, without hard pressing or toggle other complex switches, which not only simplifies the operational steps, but also improves the operational accuracy and reliability.

[0034] Further, as the preferred implementation method of this scheme rather than limited, the release mechanism **4** further includes driving parts **49** connected to the lock pins **44** through a pulling wire **48**. The connection pivot pipe **43** is provided with sliding grooves **431** for the driving part **49** to pass through and connect to the sliding sleeves **45**, and a pulling member **41** for connecting the two sliding sleeves **45**.

[0035] In this embodiment, the driving parts **49** are connected to the lock pins **44** through the pulling wire **48**. In this way, the user can control the movement of the lock pins **44** through the pulling wire **48** when operating the sliding sleeves **45**, so that the contact area between the lock pins and the inner wall of the connection pivot pipe is reduced, thereby reducing the friction. The sliding grooves **431** are provided on the connection pivot pipe **43**, providing space for the connection between the driving parts **49** and the sliding sleeves **45**. The pulling member **451** connects the two sliding sleeves **45**, so that the user can simultaneously release the lock pins **44** on

both sides with a simple pulling action, which can significantly simplify the operation process. [0036] When folding the cart, the user pulls the pulling member **451** upward, so the two sliding sleeves **45** approach each other under the pulling force. This motion is transmitted to the lock pins **44** through the driving parts **49** and the pulling wire **48**, overcoming the elastic force of the elastic member **47** and pulling out the lock pins **44** from the lock holes **46**, thereby releasing the lock of the side frame **3**. Since the pulling member **451** acts on both sliding sleeves **45** at the same time, the lock pins **44** on both sides can move synchronously, ensuring the consistency and stability of the unlocking process. This design not only avoids the trouble that the user needs to operate the lock pins on both sides separately, but also reduces the risk of unstable or accidental deployment of the cart due to asynchronous operation.

[0037] Further, as the preferred implementation method of this scheme rather than limited, when the pulling member **451** is pulled upward, both of the sliding sleeves **45** approach each other under the action of pulling force, driving the two driving parts **49** inside the connection pivot pipe **43** to pull out the corresponding lock pins **44** on both sides from the lock holes **4**, thereby releasing the restriction on the folding of the side frame **3**. Simultaneously, the side frame **3** folds and drives the front frame **1** and the rear frame **2** to approach each other, thereby completing the folding of the folding cart.

[0038] Further, as the preferred implementation method of this scheme rather than limited, the pivoting mechanism **37** is pivotally connected to the front top rod **31** through a front top pivot seat **371**, a front side of the front top pivot seat **371** is pivotally connected to the rear end of the front top rod **31** and a rear side of the front top pivot seat **371** is pivotally connected to the pivoting mechanism **37**. The pivoting mechanism **37** is pivotally connected to the rear top rod **33** through a rear top pivot seat **372**, a rear side of the rear top pivot seat **372** is pivotally connected to the front end of the rear top rod **33** and a front side of the rear top pivot seat **372** is pivotally connected to the pivoting mechanism **37**.

[0039] In this embodiment, the front top rod **31** is ensured to rotate around the pivoting mechanism **37**, so as to realize the expanding and folding of the cart. The rear side of the rear top pivot seat **372** is pivotally connected to the front end of the rear top rod **33** and the front side of the rear top pivot seat **372** is pivotally connected to the pivoting mechanism **37**, so that the rear top rod **33** can also rotate around the pivoting mechanism **37**, which not only ensures synchronized movement between the front top rod **31** and the rear top rod **33**, but also provides the essential flexibility for the folding of the side frame **3**. When the cart needs to be expanded, the front top rod **31** and the rear top rod **33** rotate forward and backward respectively under the guidance of the pivoting mechanism **37**, so that the side frame **3** is expanded and supports the entire cart. When the cart needs to be folded, the front top rod **31** and the rear top rod **33** rotate in reverse, so that the side frame **3** is folded together. Due to the existence of the front top pivot seat **371** and the rear top pivot seat **372**, this rotational motion can be carried out in a stable and controllable manner without excessive rotation or jamming.

[0040] Further, as the preferred implementation method of this scheme rather than limited, it further includes a locking mechanism **5** for keeping the folding cart in a folded state when the folding cart is folded by the side frame **3**. The locking mechanism **5** includes a folding buckle **51** and a folding hook **52** fastened on the folding buckle **51**, the folding buckle **51** is provided on the front inclined rod **35** and the folding hook **52** is provided on the rear inclined rod **36**; or the folding buckle **51** is provided on the rear inclined rod **36** and the folding hook **52** is provided on the front inclined rod **35**.

[0041] In this embodiment, the folding buckle **51** and the folding hook **52** are respectively provided on the front inclined rod **35** and the rear inclined rod **36** or respectively provided on the rear inclined rod **36** and the front inclined rod **35**. After the cart is fully folded, the folding hook **52** can accurately fastened on the folding buckle **51** to form a stable locked state. This design not only prevents the cart from accidentally expanding during carrying or storage, but also has a precise fit

and mechanical strength between the folding buckle 51 and folding hook 52. Once fastened, it can provide a reliable locking force to prevent the cart from accidentally expanding. When the cart needs to be used again, the user only needs to simply disengage the buckle between the folding hook 52 and the folding buckle 51, and the cart can be easily expanded.

[0042] Further, as the preferred implementation method of this scheme rather than limited, a slope face 441 is arranged on an end of the lock pin 44 close to the lock hole 46.

[0043] In this embodiment, the configuration of the slope face 441 can improve operational smoothness. This smooth transition not only reduces the wear and jamming among mechanical components, but also improves the operational stability and durability of the entire release mechanism 4. Especially in the case of frequent use of the cart, this design can significantly prolong the service life of the lock pins 44 and lock holes 46, and reduce the frequency of maintenance and replacement.

[0044] Further, as the preferred implementation method of this scheme rather than limited, the connection pivot pipe 43 has a prismatic shape, and the driving part 49 and the lock pin 44 are correspondingly configured as prismatic shapes.

[0045] In this embodiment, the prismatic design provides a stable and accurate motion guidance for the driving parts 49 and the lock pins 44. The corresponding prismatic configuration of the driving parts 49 and lock pins 44 enables them to closely match with the connection pivot pipe 43 and move along a predetermined trajectory, which not only prevents rotational deviation of the driving parts 49 and lock pins 44 during movement, but also ensures accurate transmission of force and motion, thereby achieving accurate control of the folding state of the side frame 3.

[0046] Further, as the preferred implementation method of this scheme rather than limited, the front top pivot seat 371 is provided with a front support rod 38 extending downward to be pivotally connected to the front bottom rod 32, and the rear top pivot seat 372 is provided with a rear support rod 39 extending downward to be pivotally connected to the rear bottom rod 39.

[0047] In this embodiment, an additional supporting frame is formed to work in concert with the original side frame 3. When the cart is in the expanded state, this design can effectively distribute loads and reduce frame deformation or shaking caused by single point stress, thereby enhancing the overall stability of the cart. This additional supporting frame can significantly enhance the cart's load-bearing capacity and shock resistance performance, especially when carrying heavy loads or operating on uneven load surfaces.

[0048] The working principle of this embodiment is as follows.

[0049] In the folding cart of the present invention, after an external force is applied to trigger the release mechanism to release the folding restriction, the external force in the same direction can be continuously applied to directly drive the side frame to fold. The user only needs to apply force in a single direction to complete the dual tasks of unlocking and folding one after another, without additionally applying forces in other directions to drive the cart to fold.

[0050] The above described is the implementation method provided in combination with the specific contents. It is not determined that the specific embodiments of the present invention are limited to these descriptions. Any methods and structures that are similar to the present invention, or any technical deductions or substitutions made under the premises of the present invention, should be regarded as the protection scope of the present invention.

Claims

1. A folding cart, comprising a front frame, a rear frame, a side frame for expanding and folding arranged between the front frame and the rear frame, and a release mechanism used for restricting or releasing restriction on the folding of the side frame; wherein a force direction applied to trigger the release mechanism to release restriction is the same as that applied to fold the side frame, after the release mechanism releases restriction on the folding of the side frame, the side frame is driven

to fold.

2. The folding cart according to claim 1, wherein the side frame includes a front top rod and a front bottom rod, whose front ends are pivotally connected to an upper side and a lower side of the front frame, respectively; a rear top rod and a rear bottom rod, whose rear ends are pivotally connected to an upper side and a lower side of the rear frame, respectively; a front inclined rod, whose front end is pivotally connected to the lower side of the front frame; a rear inclined rod, whose rear end is pivotally connected to the lower side of the rear frame; and a pivoting mechanism, which is respectively connected to a rear end of the front top rod, a front end of the rear top rod, a rear end of the front inclined rod and a front end of the rear inclined rod, and enables the four ends to be pivotally connected to each other; wherein a rear end of the front bottom rod is pivotally connected to a front end of the rear bottom rod.

3. The folding cart according to claim 2, wherein the release mechanism includes a front bottom pivot member provided at the rear end of the front bottom rod; the front end of the rear bottom rod is provided with a rear bottom pivot seat pivotally connected to a middle portion of the front bottom pivot member, a connection pivot pipe provided between the two adjacent front bottom pivot members; lock pins are configured internally in the connection pivot pipe corresponding to the connection pivot pipes on both sides, respectively, an outer side of the connection pivot pipe is sleeved with sliding sleeves connected to the lock pins; the rear bottom rod is provided with a lock hole for an insertion of the lock pins to restrict the folding of the side frame, an elastic member is provided between the two lock pins; when the folding cart is in the expanded state, the front bottom pivot member abuts against the rear bottom rod, and the elastic member drives the lock pin to insert into the lock hole.

4. The folding cart according to claim 3, wherein the release mechanism further includes driving parts connected to the lock pins through a pulling wire, the connection pivot pipe is provided with sliding grooves for the driving parts to pass through and connect to the sliding sleeves, and a pulling member connecting the two sliding sleeves.

5. The folding cart according to the claim 4, wherein when the pulling member is pulled upward, both of the sliding sleeves approach each other under a pulling force, driving the two driving parts inside the connection pivot pipe to pull out the corresponding lock pins on both sides from the lock holes, thus to release the restriction on the folding of the side frame; simultaneously, the side frame folds and drives the front frame and the rear frame to approach each other, thereby completing the folding of the folding cart.

6. The folding cart according to claim 2, wherein the pivoting mechanism is pivotally connected to the front top rod through a front top pivot seat, a front side of the front top pivot seat is pivotally connected to the rear end of the front top rod, a rear side of the front top pivot seat is pivotally connected to the pivoting mechanism; and the pivoting mechanism is pivotally connected to the rear top rod through a rear top pivot seat, a rear side of the rear top pivot seat is pivotally connected to the front end of the rear top rod, a front side of the rear top pivot seat is pivotally connected to the pivoting mechanism.

7. The folding cart according to claim 2, wherein the folding cart further includes a locking mechanism for keeping the folding cart in a folded state when the folding cart is folded, the locking mechanism includes a folding buckle and a folding hook fastened on the folding buckle, the folding buckle is provided on the front inclined rod and the folding hook is provided on the rear inclined rod; or the folding buckle is provided on the rear inclined rod and the folding hook is provided on the front inclined rod.

8. The folding cart according to claim 3, wherein a slope face is arranged on an end of the lock pin close the lock hole.

9. The folding cart according to claim 4, wherein the connection pivot pipe has a prismatic shape, and the driving part and the lock pin are correspondingly configured as prismatic shapes.

10. The folding cart according to claim 6, wherein a front support rod is configured at the front top

pivot seat, extending downwards to be pivotally connected to the front bottom rod, and a rear support rod is configured at the rear top pivot seat, extending downwards to pivotally connected to the rear bottom rod.
